Zhuchen Shao

Curriculum Vitae

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Education

University of Illinois Urbana-Champaign Sep. 2024 – May 2028 (expected)
Ph.D. Candidate in Electrical & Computer Engineering GPA: 3.73 / 4.0
Advisor: Prof. Klara Nahrstedt

Tsinghua University Sep. 2021 – Jun. 2024
M.E. in Artificial Intelligence GPA: 3.63 / 4.0
Advisor: Prof. Haoqian Wang

Dalian Maritime University Sep. 2017 – Jun. 2021
B.E. in Automation GPA: 4.22 / 5.0 Rank: 1 of 123

Academic Research

Embodied Multi-Agent Planning Ongoing

- Developing and testing planners for embodied multi-agent collaboration on the PARTNR benchmark.

Streaming Video Understanding

- Developed StreamClarify, a training-free streaming video QA framework for frozen vision-language models (VLMs), using entropy-based clarification, temporal-pyramid key-value caching, and semantic retrieval.
- Built StreamClarify-QA with 1,837 QA pairs from 355 videos and evaluated three VLM backbones. Improved macro accuracy by up to 8.2 percentage points over existing streaming video QA methods.

Computer Vision for Medical Imaging

- Developed Transformer-based multiple instance learning methods for weakly supervised whole-slide image classification and survival prediction, using spatial context and noisy pseudo-labels.
- Applied diffusion models to feature augmentation and semi-supervised cell segmentation, using unsupervised pre-training, collaborative learning, and content-guided sampling.
- Studied task-informed image reconstruction with a data-consistent U-Net and null-space learning under task and observer shifts.

Industry Research

Applied Scientist Intern · Amazon Alexa AI (Multimodal Agents) May 2026 – Aug. 2026

- Developed Tab Zero, a unified browser interface that lets web agents inspect egocentric video and complete web tasks using shared browser actions.
- Built and tested a distributed reinforcement learning pipeline with GRPO, SGLang, Megatron-LM, and parallel Chromium rollouts. Tested rule-based and VLM-based rewards.
- Evaluated three model families on Ego2Web and LongEval (3–20-minute videos). Improved zero-shot task success rate by up to 8.0 percentage points over uniform sampling on the filtered Ego2Web subset.

Honors & Skills

National Scholarship (top 1%), Tsinghua University 2023

National Scholarship (top 1%), Dalian Maritime University 2018, 2019, 2020

Programming & Tools: Python, PyTorch, LaTeX

Languages: English (fluent), Mandarin (native)

Publications

Streaming Video Understanding & Multimodal Agents

1. **StreamClarify: Proactive Clarification for Grounded Streaming Video Question Answering**. **Zhuchen Shao**, Bo Chen, Klara Nahrstedt. *WACV, 2027 (accepted)*
2. **Tab Zero: Grounding Web Agents in Egocentric Video**. **Zhuchen Shao**, et al. *In preparation*

Computer Vision for Medical Imaging

3. **Investigating the impact of data consistency in task-informed learned image reconstruction method**. **Zhuchen Shao**, Changjie Lu, Kaiyan Li, Hua Li, Mark A. Anastasio. *SPIE Medical Imaging, 2025* [Paper]
4. **Prior-guided diffusion model for cell-level segmentation in quantitative phase imaging**. **Zhuchen Shao**, Mark A. Anastasio, Hua Li. *SPIE Medical Imaging, 2025* [Paper]
5. **Semi-supervised semantic segmentation of cell nuclei with diffusion model and collaborative learning**. **Zhuchen Shao**, Sourya Sengupta, Mark A. Anastasio, Hua Li. *Journal of Medical Imaging, 2025* [Paper]
6. **AugDiff: Diffusion based Feature Augmentation for Multiple Instance Learning in Whole Slide Image**. **Zhuchen Shao**, Liuxi Dai, Yifeng Wang, Haoqian Wang, Yongbing Zhang. *IEEE Transactions on Artificial Intelligence, 2024* [Paper]
7. **LNPL-MIL: Learning from Noisy Pseudo Labels for Promoting Multiple Instance Learning in Whole Slide Image**. **Zhuchen Shao**, Yifeng Wang, Yang Chen, Hao Bian, Shaohui Liu, Haoqian Wang, Yongbing Zhang. *ICCV, 2023* [Paper]
8. **HVTSurv: Hierarchical Vision Transformer for Patient-level Survival Prediction from Whole Slide Image**. **Zhuchen Shao**, Yang Chen, Hao Bian, Jian Zhang, Guojun Liu, Yongbing Zhang. *AAAI, 2023* [Paper]
9. **dMIL-Transformer: Multiple Instance Learning via Integrating Morphological and Spatial Information for Lymph Node Metastasis Classification**. Yang Chen*, **Zhuchen Shao***, Hao Bian*, Zijie Fang, Yifeng Wang, Yuanhao Cai, Haoqian Wang, Guojun Liu, Xi Li, Yongbing Zhang. *IEEE Journal of Biomedical and Health Informatics, 2023* [Paper]
10. **Multiple Instance Learning with Mixed Supervision in Gleason Grading**. Hao Bian*, **Zhuchen Shao***, Yang Chen*, Yifeng Wang, Haoqian Wang, Jian Zhang, Yongbing Zhang. *MICCAI, 2022* [Paper]
11. **TransMIL: Transformer based Correlated Multiple Instance Learning for Whole Slide Image Classification**. **Zhuchen Shao***, Hao Bian*, Yang Chen*, Yifeng Wang, Jian Zhang, Xiangyang Ji, Yongbing Zhang. *NeurIPS, 2021* [Paper]

* Equal contribution.